

Overview of the Iterative Proportional Fitting

Let $\mathcal{P}_l = \mathcal{P}((\mathbb{R}^d)^l)$ be the space of all probability measures on $(\mathbb{R}^d)^l$. IPF is defined by the following recursion for $n \in \mathbb{N}$:

$$\pi^{2n+1} = \operatorname{argmin}\{\operatorname{KL}(\pi \mid \pi^{2n}) : \pi \in \mathcal{P}_{N+1}, \pi_N = p_{\text{prior}}\} \quad (1)$$

$$\pi^{2n+2} = \operatorname{argmin}\{\operatorname{KL}(\pi \mid \pi^{2n+1}) : \pi \in \mathcal{P}_{N+1}, \pi_0 = p_{\text{data}}\} \quad (2)$$

with initialization $\pi^0 = p$, and $p_{\text{data}}, p_{\text{prior}}$ as the prescribed marginals. Under the assumptions laid out in [Diffusion Schrödinger Bridge with Applications to Score-Based Generative Modeling](#), the sequence $(\pi^n)_{n \in \mathbb{N}}$ is well defined. It also follows that for any $n \in \mathbb{N}$:

$$\operatorname{KL}(\pi^{n+1} \mid \pi^n) \leq \operatorname{KL}(\pi^{n-1} \mid \pi^n), \quad \operatorname{KL}(\pi^n \mid \pi^{n+1}) \leq \operatorname{KL}(\pi^n \mid \pi^{n-1})$$

and the errors $\operatorname{KL}(\pi_0^n \mid p_{\text{data}}), \operatorname{KL}(\pi_N^n \mid p_{\text{prior}})$ shrink proportionally to $O(1/n)$:

$$\lim_{n \rightarrow \infty} n[\operatorname{KL}(\pi_0^n \mid p_{\text{data}}) + \operatorname{KL}(\pi_N^n \mid p_{\text{prior}})] = 0$$

It can also be shown that if $\pi^* \in \mathcal{P}_{N+1}$ the solution of the Schrödinger Problem, if the IPF iterations converge to $\pi^\infty \in \mathcal{P}_{N+1}$, meaning that $\lim_{n \rightarrow \infty} \|\pi^n - \pi^\infty\|_{\text{TV}} = 0$, and if the initialization p is absolute continuous to the limit π^∞ then the IPF iterations converges to the Schrödinger Bridge:

$$\pi^\infty = \pi^*$$

1 Diffusion Schrödinger Bridge

Assume that we are at the step $2n+1$, for $n \in \mathbb{N}$, and let π^{2n+1} be the optimal measure according to the backward IPF iteration (1). From the Entropic Decomposition Lemma, we can write the KL divergence between $\pi \in \mathcal{P}_{N+1}$ and π^{2n} as:

$$\operatorname{KL}(\pi \mid \pi^{2n}) = \operatorname{KL}(\pi_N \mid \pi_N^{2n}) + \int_{\mathbb{R}^d} \operatorname{KL}(\pi_{|N}(\cdot \mid z) \mid \pi_{|N}^{2n}(\cdot \mid z)) \pi_N(z) dz$$

where we denote π_N as the marginal distribution of π at time step N , and $\pi_{|N}(\cdot \mid z)$ as the conditional distribution of the path given the state z at time step N .

At the minimization point we have $\pi = \pi^{2n+1}$, which means that:

$$\pi_N^{2n+1} = p_{\text{prior}}, \quad \pi_{|N}^{2n+1} = \pi_{|N}^{2n}$$

Similarly we can show that from the $2n + 2$ step we have:

$$\pi_0^{2n+2} = p_{\text{data}}, \quad \pi_{|0}^{2n+1} = \pi_{|0}^{2n+1}$$

which means that if π^{2n} admits a density p^n with respect to the Lebesgue measure, and π^{2n+1} admits q^n , then they are expressed as:

$$q^n(x_{0:N}) = p_{\text{prior}}(x_N) \prod_{k=0}^{N-1} p_{k|k+1}^n(x_k | x_{k+1})$$

$$p^{n+1}(x_{0:N}) = p_{\text{data}}(x_0) \prod_{k=0}^{N-1} q_{k+1|k}^n(x_{k+1} | x_k)$$

Let $(X_t)_{t \in [0, T]}$ be the stochastic process whose law is the forward path measure p^n , and assume it satisfies the following SDE:

$$dX_t = f_t^n(X_t)dt + dB_t$$

Then from the Euler-Maruyama discretization we derive that:

$$p_{k+1|k}^n(x_{k+1} | x_k) = N(x_{k+1} ; x_k + \gamma_{k+1} f_k^n(x_k), \gamma_{k+1} \mathbf{I})$$

The reverse-time transitions $q_{k|k+1}^n$ are intractable because they require the calculation of the marginals p_k^n and p_{k+1}^n using Bayes' rule:

$$q_{k|k+1}^n(x_k | x_{k+1}) = p_{k+1|k}^n(x_{k+1} | x_k) = \frac{p_{k+1|k}^n(x_{k+1} | x_k) p_k^n(x_k)}{p_{k+1}^n(x_{k+1})}$$

We can rewrite the reverse-time transitions as a logarithmic difference of the intractable parts:

$$q_{k|k+1}^n(x_k | x_{k+1}) = p_{k+1|k}^n(x_{k+1} | x_k) \exp\{\log p_k^n(x_k) - \log p_{k+1}^n(x_{k+1})\}$$

Under mild assumptions which are justified by the small step size γ_{k+1} it can be shown (for the exact derivation see [Diffusion Schrödinger Bridge with Applications to Score-Based Generative Modeling](#)):

$$q_{k|k+1}^n(x_k | x_{k+1}) \approx N(x_k ; x_{k+1} + \gamma_{k+1} b_{k+1}^n(x_{k+1}), \gamma_{k+1} \mathbf{I})$$

where $b_{k+1}^n(x_{k+1}) = -f_k^n(x_{k+1}) + \nabla \log p_{k+1}^n(x_{k+1})$.

Similarly, to derive the forward density of the next step $p_{k+1|k}^{n+1}$, we obtain:

$$p_{k+1|k}^{n+1}(x_{k+1} | x_k) \approx N(x_{k+1} ; x_k + f_k^{n+1}(x_k), \gamma_{k+1} \mathbf{I})$$

where $f_k^{n+1}(x_k) = -b_{k+1}^n(x_k) + \nabla \log q_k^n(x_k)$.

These approximations allow us to estimate the forward and backward transitions as follows. Let the drifts of the two processes be parameterized by the functions $B_{k+1}^n(x) = x + \gamma_{k+1}b_{k+1}^n(x)$ and $F_k^n(x) = x + \gamma_{k+1}f_k^n(x)$ for any $x \in \mathbb{R}^d$. Then we can write the transitions as:

$$\begin{aligned} q_{k|k+1}^n(x_k | x_{k+1}) &= \mathcal{N}(x_k ; B_{k+1}^n(x_{k+1}), \gamma_{k+1}\mathbf{I}) \\ p_{k+1|k}^{n+1}(x_{k+1} | x_k) &= \mathcal{N}(x_{k+1} ; F_k^{n+1}(x_k), \gamma_{k+1}\mathbf{I}) \end{aligned}$$

and for any $n \in \mathbb{N}$ and $k \in \{0, 1, \dots, N-1\}$:

$$\begin{aligned} B_{k+1}^n &= \arg \min_{B \in L^2(\mathbb{R}^d)} \mathbb{E}_{(x_k, x_{k+1}) \sim p_{k,k+1}^n} [\|B(x_{k+1}) - (x_{k+1} + F_k^n(x_k) - F_k^n(x_{k+1}))\|^2] \\ F_k^{n+1} &= \arg \min_{F \in L^2(\mathbb{R}^d)} \mathbb{E}_{(x_k, x_{k+1}) \sim q_{k,k+1}^n} [\|F(x_k) - (x_k + B_{k+1}^n(x_{k+1}) - B_{k+1}^n(x_k))\|^2] \end{aligned}$$

Finally, following [Diffusion Schrödinger Bridge with Applications to Score-Based Generative Modeling](#), in the continuous setting we denote $\mathcal{P}(C)$ the space of all probability measure on the path space $C([0, 1], \mathbb{R}^d)$, and the IPF iterations on the sequence $(\Pi^n)_{n \in \mathbb{N}}$ as:

$$\begin{aligned} \Pi^{2n+1} &= \operatorname{argmin}\{\text{KL}(\Pi | \Pi^{2n}) : \Pi \in \mathcal{P}(C), \Pi_1 = p_{\text{prior}}\} \\ \Pi^{2n+2} &= \operatorname{argmin}\{\text{KL}(\Pi | \Pi^{2n+1}) : \Pi \in \mathcal{P}(C), \Pi_0 = p_{\text{data}}\} \end{aligned}$$

with initialization $\Pi^0 = P$, associated with the following SDE:

$$dX_t = f(X_t)dt + dB_t$$

The convergence results can also be expanded for the continuous setting. Define P^R the time-reversal of P . Then under mild assumptions we can show that $(\Pi^{2n+1})^R$ is associated with:

$$dY_t^{2n+1} = b_{1-t}^n(Y_t^{2n+1})dt + dB_t, \quad Y_0^{2n+1} \sim p_{\text{prior}}$$

Furthermore, Π^{2n+2} is associated with:

$$dX_t^{2n+1} = f_t^{n+1}(X_t^{2n+2})dt + dB_t, \quad X_0^{2n+2} \sim p_{\text{data}}$$

where for any $n \in \mathbb{N}$, $t \in [0, 1]$, and $x \in \mathbb{R}^d$ we have:

$$\begin{aligned} b_t^n(x) &= -f_t^n(x) + \nabla \log p_t^n(x) \\ f_t^{n+1}(x) &= -b_t^n(x) + \nabla \log q_t^n(x) \end{aligned}$$

where $f_t^0(x) = f(x)$, and p_t^n , q_t^n the density of Π_t^{2n} , Π_t^{2n+1} respectively. Which shows that SDB is the discretization of IPF.

Algorithm 1:

Input: Data distribution p_{data} , Prior distribution p_{prior} , Number of time steps $N \in \mathbb{N}$, Step sizes $\{\gamma_k\}_{k=1}^N$, Number of outer iterations $M \in \mathbb{N}$

Initialize forward map $F_\theta(k, \cdot)$ to match a reference process (e.g., $F_\theta(k, x) = x$).

for $n = 0$ **to** $M - 1$ **do**

Step 1: Generate forward trajectories p^n starting from the data ($x_0 \sim p_{\text{data}}$).

$$x_{k+1} \sim N(x_{k+1} ; F_\theta(k, x_k), \gamma_{k+1} \mathbf{I}), \quad k = 0, \dots, N - 1$$

Step 2: Fit the backward drift network B_ϕ by solving:

$$\phi^* = \operatorname{argmin}_\phi \left\{ \mathbb{E}_{k \sim U\{0, \dots, N-1\}} \mathbb{E}_{(x_k, x_{k+1}) \sim p^n} \left[\left\| B_\phi(k+1, x_{k+1}) - (x_{k+1} + F_\theta(k, x_k) - F_\theta(k, x_{k+1})) \right\|^2 \right] \right\}$$

Update backward map: $B_\phi \leftarrow B_{\phi^*}$

Step 3: Generate backward trajectories q^n starting from the prior ($x_N \sim p_{\text{prior}}$).

$$x_k \sim \mathcal{N}(x_k ; B_\phi(k+1, x_{k+1}), \gamma_{k+1} \mathbf{I}), \quad k = N - 1, \dots, 0$$

Step 4: Fit the forward drift network F_θ by solving:

$$\theta^* = \operatorname{argmin}_\theta \left\{ \mathbb{E}_{k \sim U\{0, \dots, N-1\}} \mathbb{E}_{(x_k, x_{k+1}) \sim q^n} \left[\left\| F_\theta(k, x_k) - (x_k + B_\phi(k+1, x_{k+1}) - B_\phi(k+1, x_k)) \right\|^2 \right] \right\}$$

Update forward map: $F_\theta \leftarrow F_{\theta^*}$

Output: Optimized forward map F_{θ^*} and backward map B_{ϕ^*}

Algorithmic Implementation

Steps 1,3: Simulating Trajectories to Build the Cache

Before solving the backward regression, we must generate trajectories from the current forward path measure p^n . The forward chain is initialized at the data distribution

$x_0 \sim p_{\text{data}}$ and simulated using the Euler-Maruyama discretization:

$$x_{k+1} = x_k + hf_{\theta}(x_k, t_k) + \sqrt{h}z, \quad z \sim \mathcal{N}(0, \mathbf{I})$$

For the very first IPF iteration ($n = 0$), the forward model is replaced by a reference process, typically Standard Brownian Motion where the drift evaluates to zero: $f(x, t) = 0$.

Similarly, in Step 3, before solving the forward regression, we must generate trajectories from the recently updated backward path measure q^n . The backward chain is initialized at the prior distribution $x_N \sim p_{\text{prior}}$ (typically standard Gaussian $\mathcal{N}(0, \mathbf{I})$) and simulated backward in time ($k = N-1, \dots, 0$) using its respective Euler-Maruyama discretization:

$$x_k = x_{k+1} + hb_{\phi}(t_{k+1}, x_{k+1}) + \sqrt{h}z, \quad z \sim \mathcal{N}(0, \mathbf{I})$$

By simulating and caching these transitions, we construct the empirical distributions of sequential pairs (x_k, x_{k+1}) required to approximate the expectations $\mathbb{E}_{(x_k, x_{k+1}) \sim p^n}$ and $\mathbb{E}_{(x_{k+1}, x_k) \sim q^n}$ during the regression steps.

Steps 2,4: Solving the Regression Problems

The algorithm minimizes the following mean squared error (MSE) objective:

$$\phi^* = \operatorname{argmin}_{\phi} \left\{ \mathbb{E}_{k \sim U\{0, \dots, N-1\}} \mathbb{E}_{(x_k, x_{k+1}) \sim p^n} \left[\left\| B_{\phi}(k+1, x_{k+1}) - (x_{k+1} + F_{\theta}(k, x_k) - F_{\theta}(k, x_{k+1})) \right\|^2 \right] \right\}$$

We parameterize the backward and forward transitions using neural networks b_{ϕ} and f_{θ} to model the residual drifts:

$$\begin{aligned} B_{\phi}(t_k, x_k) &= x_k + hb_{\phi}(t_k, x_k) \\ F_{\theta}(t_k, x_k) &= x_k + hf_{\theta}(t_k, x_k) \end{aligned}$$

The forward density p^n is governed by the stochastic differential equation $dX_t = f_{\theta}(t, X_t)dt + dB_t$, which yields the following Euler-Maruyama discretization:

$$x_{k+1} = x_k + hf_{\theta}(t_k, x_k) + \sqrt{h}z, \quad z \sim N(0, \mathbf{I})$$

Substituting this discretization directly into the reparameterized objective isolates the drift components, yielding the final simplified loss function:

$$\phi^* = \operatorname{argmin}_{\phi} \left\{ \mathbb{E}_{k \sim U\{0, \dots, N-1\}} \mathbb{E}_{(x_k, x_{k+1}) \sim p^n} \left[\left\| b_{\phi}(t_{k+1}, x_{k+1}) - \left(-f_{\theta}(t_{k+1}, x_{k+1}) - \frac{z}{\sqrt{h}} \right) \right\|^2 \right] \right\}$$

However, the presence of the term $z/\sqrt{h}$ implies that the variance of the target scales with $\mathcal{O}(1/h)$.

To ensure numerical stability and bound the variance to $\mathcal{O}(1)$, we scale the objective by the step size h . This yields the final scaled regression objective:

$$\phi^* = \operatorname{argmin}_{\phi} \left\{ \mathbb{E}_{k \sim U\{0, \dots, N-1\}} \mathbb{E}_{(x_k, x_{k+1}) \sim p^n} \left[h \left\| b_{\phi}(t_{k+1}, x_{k+1}) + f_{\theta}(t_{k+1}, x_{k+1}) + \frac{z}{\sqrt{h}} \right\|^2 \right] \right\}$$

Similarly, for Step 4, substituting the backward trajectory discretization $x_k = x_{k+1} + hb_{\phi}(t_{k+1}, x_{k+1}) + \sqrt{h}z$ into the reparameterized objective yields the drift-level target for the forward model. Applying the same variance-stabilizing scaling factor h , we obtain the final regression problem:

$$\theta^* = \operatorname{argmin}_{\theta} \left\{ \mathbb{E}_{k \sim U\{0, \dots, N-1\}} \mathbb{E}_{(x_{k+1}, x_k) \sim q^n} \left[h \left\| f_{\theta}(t_k, x_k) + b_{\phi}(t_{k+1}, x_k) + \frac{z}{\sqrt{h}} \right\|^2 \right] \right\}$$